MILESTONE P2 — WEEK 7

ACADEMIC YEAR 2026–2027

E-Commerce Semester Project

Specification & Working Prototype

Frozen technical baseline, architectural models, and functional catalog demo.

BACKEND & DB LEAD

Necșulea Andrei-Mihai

FRONTEND & UI/UX LEAD

Onofrei Alexandru-Gabriel

STATUS

Specification Frozen

P2 EXECUTIVE OVERVIEW

Store Scope & Target Strategy

PRODUCT NICHE

Dual PC & Industrial Automation Catalog

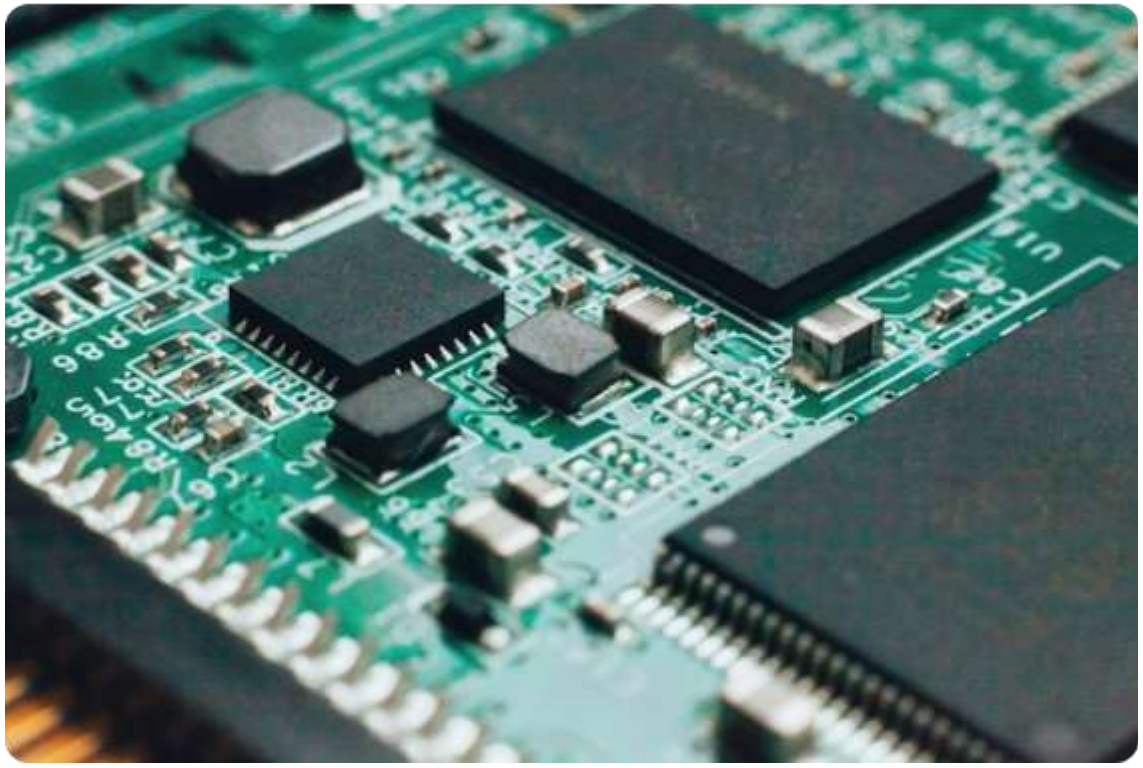
Bridging high-end personal computing components with industrial PLCs, I/O modules, and automation hardware in one unified platform.

B2C Segment

Engineers and builders buying CPUs, RAM, SSDs, and test bench gear.

B2B Segment

Workshops purchasing PLCs and automation modules without quote lag.



IMPLEMENTATION STRATEGY 1/3

Requirements A1 to A4: Core Catalog & Users

A1

Product Catalog

Django views with built-in Paginator (max 24 items/page), category filtering, and detailed product cards with live stock.

Owner: Andrei & Alexandru

A2

Basic Search

Case-insensitive contains query across product titles and SKUs, rendering structured empty-state templates when no match exists.

Owner: Alexandru

A3

Shopping Cart

Session cart for guests merged into relational database on login; server strictly recalculates totals and validates stock.

Owner: Andrei

A4

User Accounts

Standard auth with Argon2 password hashing, profile management, and relational address storage with default billing/shipping toggles.

Owner: Andrei

IMPLEMENTATION STRATEGY 2/3

Requirements A5 to A8: Checkout, Payment & Stock

A5

Multi-step Checkout

Guided four-step flow: address confirmation, courier choice (Express vs Standard), order breakdown, and final confirmation view.

Owner: Alexandru

A6

Sandbox Payment

Stripe checkout session; order transitions to Paid strictly upon signature-verified webhook receipt, never browser redirection.

Owner: Andrei

A7

Stock Management

PostgreSQL row locking (select_for_update) decrements inventory atomically on webhook, preventing overselling the final unit.

Owner: Andrei

A8

Order History

Authenticated dashboard listing past orders with timestamps, unit line items, total pricing, and real-time shipment status badges.

Owner: Alexandru

IMPLEMENTATION STRATEGY 3/3

Requirements A9 to A12 & Targeted Bonuses

A9 — Custom Admin Panel

Hand-crafted Django admin views for product/category CRUD and order transitions: New → Paid → Shipped → Delivered.

A10 — Robust Security

CSRF middleware, parameterized ORM calls, and server-side role decorators strictly forbidding non-staff access.

A11 — Responsive UI (360px)

Tailwind mobile-first grid and flex layouts verified across standard mobile viewports down to 360px viewport width.

A12 — Automated Tests

Suite of 12 Django TestCase units covering cart calculations, stock boundaries, and webhook events with one command.

APPROVED BONUS FEATURES SCOPE (TARGETING MAX 20 PTS)

B1: Filtered Search (+5 pts)

Combinable category, price range, and stock filters with preserved URL queries.

B5: Discount Codes (+3 pts)

Server-validated promo codes checking expiry dates and maximum usage quotas.

B6: Transactional Mail (+5 pts)

HTML order receipts and shipment updates dispatched via Mailtrap SMTP.

B7: Wishlist (+2 pts)

Authenticated users can save preferred hardware or PLC modules to a persistent wishlist with a single click, monitor stock availability, and seamlessly transfer items directly to the shopping cart when ready to purchase.

RELATIONAL SCHEMA

Entity-Relationship Data Model

User & Address

User
- id (PK, UUID)
- email (Unique)
- password_hash
- is_staff (Boolean)

Address
- id (PK)
- user_id (FK)
- street, city, postal
- is_default (Bool)

Catalog Entities

Category
- id (PK)
- name, slug (Unique)
- description

Product
- id (PK, UUID)
- category_id (FK)
- title, sku, specs
- price (Decimal)
- stock_quantity (Int)
- image_url

Cart Subsystem

Cart
- id (PK)
- user_id (FK, Nullable)
- session_key (Index)
- updated_at

CartItem
- id (PK)
- cart_id (FK)
- product_id (FK)
- quantity (Int)

Order & Payment

Order
- id (PK, UUID)
- user_id (FK)
- status (Enum)
- total_amount
- stripe_session_id

OrderItem
- id (PK)
- order_id (FK)
- product_id (FK)
- unit_price, quantity

SYSTEM DESIGN

Component Diagram & Layer Interaction

CLIENT TIER

Web Browser / Mobile

Responsive Tailwind templates, HTML5 forms & Stripe Checkout SDK

APPLICATION ENGINE

Django Core Framework

Middleware, Routing, View Controllers, ORM, Auth & Custom Admin

PERSISTENCE

PostgreSQL 16

ACID storage, Row-level Locking, Migrations & Seed Datasets

Third-Party Integrations Isolation

Stripe Sandbox for PCI-compliant checkout sessions + Mailtrap for outbound transactional SMTP dispatch.

Docker Orchestrated

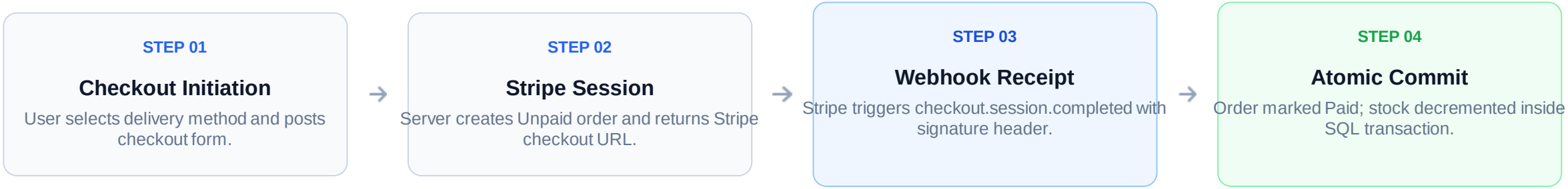
INTERFACE CONTRACTS

API & Route Endpoints Matrix

METHOD	ROUTE	ACCESS	PURPOSE
GET	/catalog/?page=1&cat=plc	Public	Paginated catalog & filter execution (A1, B1)
POST	/cart/add//	Public / Session	Add item & validate server-side stock quota (A3)
POST	/checkout/create-session/	**Authenticated**	Initialize Stripe checkout session (A5, A6)
POST	/api/stripe/webhook/	**Stripe Signature**	Confirm payment & trigger atomic stock reduction (A6, A7)
GET	/orders/history/	**Authenticated**	Render customer order logs and tracking states (A8)
ALL	/portal-admin/*	**Staff / Admin**	Custom admin views for products & orders lifecycle (A9)

CRITICAL PATH SPECIFICATION

Asynchronous Payment & Stock Sequence Flow



Strict Specification Compliance Check

Browser redirection lands on a pending confirmation screen. Order status is never flipped to Paid until the Stripe webhook signature is verified on the backend.

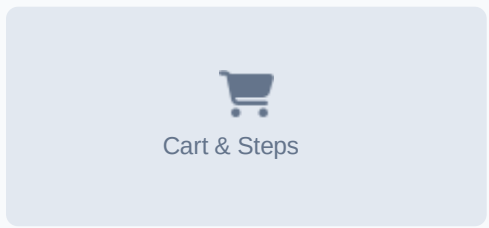
VISUAL ARCHITECTURE

Key Interface Wireframes & Layout Grid



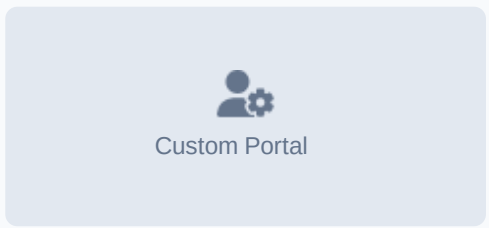
Catalog & Product Page

Sidebar categories, sticky search bar, live stock counters, and responsive product cards.



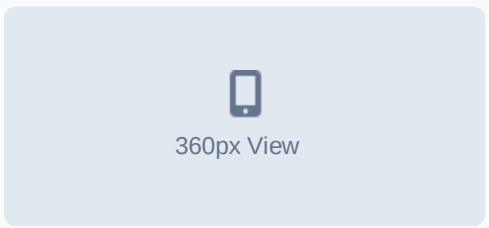
Cart & Checkout Flow

Dynamic row updates, stock alerts, courier radio selector, and Stripe embedded payment.



Custom Admin Panel

Product CRUD modals, direct media upload handling, and one-click order status update transitions.



Mobile Responsive Pass

Collapsible navigation drawer, touch-friendly CTA buttons, and vertical summary stacking.

P2 LIVE DEMONSTRATION

Working Prototype Readiness & Datasets

36	4	100%
Seeded Products (A1/Data)	Distinct Categories	Live Database Driven

Live Demo Sequence for P2 Evaluation:

1. User registers new account & logs in → 2. Catalog renders real seeded PostgreSQL records with pagination → 3. Execute search query by item name and show zero-lag filtered responses.